

Quiz 1

NAME: _____

Q.1. [5 points] Simplify the following expression:

$$\sqrt[3]{3a^2b^3} \cdot \sqrt[3]{6a^5b}$$

Q.2. [5 points] Simplify the following expression:

$$\left[\frac{x-2y}{5z+1} \right] \div \left[\frac{3x-6y}{-z-(2+9z)} \right]$$

Q.3. [5 points] Fully factor the following polynomial:

$$x^2y + 3x^2 - 4y - 12$$

Q.4. [5 points] Fully factor the following polynomial:

$$-3x^2 - 4x + 15$$

BONUS Simplify the following expression (a, b, c can be any real numbers here):

$$\frac{\sqrt[3]{a^6 \cdot b^{-3}}}{\sqrt{a^2 \cdot c^8}}$$